

MATH 231 – Linear Algebra I

The Matrix Unit, Part 1 ■ Linear Systems and the Matrix

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About this document. This opens the unit on *matrices*. It does not begin by defining one. It begins with a question the vector unit left unanswered — given a basis and a vector, how do you find the coordinates? — shows that the question is a system of n linear equations, and then builds the matrix as the notation that system demands.

A common confusion (Section numbers restart). This unit numbers its sections from 1 again. They do *not* continue from the vector unit, whose sections run §1 to §24, and they have nothing to do with the computational unit, which also restarts at 1. So a bare reference such as “§1.3” below means §1.3 *of this unit*. Whenever we point at either of the other two we will name it, as in “the vector unit, §16.5”.

Assumed from the vector unit. $\mathbb{R}^n$ and the bracket notation $\mathbf{v} = [v_1, \dots, v_n]$ (§24.1); addition and scalar multiplication (§7–§8); the dot product $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_i u_i v_i$ (§11.3); linear combinations and span (§16.2–§16.3); coordinates and their uniqueness (§16.4); the coordinate problem for the standard basis (§16.5); linear independence (§18.4); basis and dimension (§16.3, §17); and orthogonality (§11.7).

What you should be able to do afterwards. Say why a basis turns “find the coordinates of $\mathbf{u}$ ” into a system of n linear equations in n unknowns; write that system out from a given basis; say what it means to solve it; recognise each equation as a dot product; build the coefficient matrix $\mathbf{V}$ and the coordinate vector $\mathbf{x}$; state the definition of a matrix–vector product and compute one; write a system as $\mathbf{V}\mathbf{x} = \mathbf{u}$; and read that single equation in both directions — by rows as n dot products, and by columns as one linear combination. Then: define a matrix and address its entries; name the standard shapes and patterns; add, scale and transpose matrices and recognise symmetry; write the dot product, the outer product and the norm in transpose notation; multiply two matrices and check in advance whether the product exists; and compute the trace of a square matrix.

1. The problem a basis leaves behind

1.1. What a basis promises

Work in $\mathbb{R}^n$ and let

$$\mathcal{B} = \{ \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \}$$

be a set of n linearly independent vectors. That is enough to make $\mathcal{B}$ a basis.

From the vector unit. Two facts do the work. A basis for $\mathbb{R}^n$ is a set that spans $\mathbb{R}^n$ and is linearly independent (§16.3), and a basis for $\mathbb{R}^n$ has exactly n vectors, since $\dim \mathbb{R}^n = n$ (§17, §24.1). What is not obvious is that independence alone is enough — that n independent vectors cannot fail to span. The vector unit proved this in $\mathbb{R}^2$ (§18.5) and used it in $\mathbb{R}^n$ (§24.4). We take it as given.

Remark. It is worth being clear that this is a genuine theorem and not a definition. In principle a set could be independent and still miss part of the space; what the result says is that in $\mathbb{R}^n$, once you have n independent vectors, there is no room left for it to miss anything. A proof in general dimension is one of the things the machinery of this unit will eventually supply, which is a fair reason to postpone it rather than assert it twice.

Being a basis, $\mathcal{B}$ spans $\mathbb{R}^n$, and *span* means exactly what it said in the vector unit, §16.3: the set of all linear combinations. So every vector of $\mathbb{R}^n$ is reachable. Written out: for any $\mathbf{u} \in \mathbb{R}^n$ there are scalars $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ with

$$\mathbf{u} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n = \sum_{i=1}^n \alpha_i \mathbf{v}_i$$

and, because coordinates against a basis are unique (§16.4), there is exactly one such list of scalars.

That is a promise about existence, and it is the kind of promise that is easy to undervalue. It says: pick any point of $\mathbb{R}^n$ you like — any point at all, however far out, in however many dimensions — and it can be built out of these n vectors, in one and only one way.

What the promise does *not* do is tell you the scalars.

1.2. The reverse problem

So we ask for them. This is the reverse problem of the vector unit, §16.5, now posed in $\mathbb{R}^n$ and against an arbitrary basis rather than the standard one:

Given a basis $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and a vector $\mathbf{u} \in \mathbb{R}^n$, find the scalars $\alpha_1, \dots, \alpha_n$ with $\mathbf{u} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n$.

The $\mathbf{v}_i$ are known. The vector $\mathbf{u}$ is known. The α_i are the unknowns, and §1.1 has already told us two things about them before we begin: they exist, and there is only one list of them. We are not searching hopefully. We are computing something we know is there and know is unambiguous — which is precisely the situation in which it is worth developing a method.

From the vector unit. When $\mathcal{B}$ was the standard basis this question was free. Because $\mathbf{e}_i$ has a 1 in slot i and zeros elsewhere, the coordinates of $\mathbf{u}$ are just its components, $\alpha_i = u_i$, read off with no work at all (§16.5 in $\mathbb{R}^2$, §24.1 in $\mathbb{R}^n$). The vector unit was careful to say why: the standard basis is orthonormal, and the zeros that make the reading free are the statement $\langle \mathbf{e}_i, \mathbf{e}_j \rangle = 0$. Change the basis and the zeros go, and with them the free answer.

1.3. The system, written out

To turn the vector equation into something we can compute with, expand it componentwise. Two vectors in $\mathbb{R}^n$ are equal exactly when all n of their components agree (§9), so one vector equation is n scalar equations.

First, notation for the components of the basis vectors. Each $\mathbf{v}_i$ is a vector in $\mathbb{R}^n$, so it has n components, and we need a symbol carrying both *which vector* and *which slot*:

$$\mathbf{v}_i = [v_{1i}, v_{2i}, \dots, v_{ni}], \quad \text{so} \quad v_{ki} = \text{the } k\text{-th component of } \mathbf{v}_i.$$

A common confusion (Which index is which). Read v_{ki} as “slot k , vector i ”. The *first* index says which component and the *second* says which basis vector — the opposite of the order in which you would say it out loud. This is deliberate, and §2.2 is where it pays: written this way, v_{ki} will turn out to sit in row k and column i of the object we are about to build, so the two indices are the two directions of a grid, in that order. Choose the other convention and you spend the rest of the unit transposing in your head.

Now expand. Scaling $\mathbf{v}_i$ by α_i multiplies every component (§8), and adding vectors adds them slot by slot (§7), so slot k of the right-hand side collects one term from each basis vector:

$$\text{slot } k \text{ of } \sum_{i=1}^n \alpha_i \mathbf{v}_i = \alpha_1 v_{k1} + \alpha_2 v_{k2} + \cdots + \alpha_n v_{kn}.$$

Setting that equal to u_k for each $k = 1, \dots, n$ gives n equations:

$$\begin{aligned} \alpha_1 v_{11} + \alpha_2 v_{12} + \cdots + \alpha_n v_{1n} &= u_1, \\ \alpha_1 v_{21} + \alpha_2 v_{22} + \cdots + \alpha_n v_{2n} &= u_2, \\ &\vdots \\ \alpha_1 v_{n1} + \alpha_2 v_{n2} + \cdots + \alpha_n v_{nn} &= u_n. \end{aligned}$$

Definition 1.1 (Linear system). A **linear system** is a finite collection of linear equations in the same unknowns, to be satisfied *simultaneously*. The display above is a system of n equations in the n unknowns $\alpha_1, \dots, \alpha_n$.

Read the display in both directions once, because both readings are used constantly.

- **Across a row.** Row k is one equation. It involves every unknown, and the coefficients it uses are the k -th components of all n basis vectors.
- **Down a column.** Column i belongs to one basis vector. Reading $v_{1i}, v_{2i}, \dots, v_{ni}$ downward reassembles $\mathbf{v}_i$ itself.

That the same array of numbers organises into rows that are equations and columns that are vectors is not a coincidence, and it is the whole reason §2 works.

1.4. What it means to solve it

Definition 1.2 (Solving a linear system). To **solve** a linear system is to find values of the unknowns that satisfy *every* equation at once. A list of values satisfying all of them is a **solution**.

The word *simultaneously* is the entire difficulty. Any one of those n equations is easy on its own — one equation in n unknowns has endlessly many solutions, and you can satisfy it by picking $n - 1$ of the α 's at random and solving for the last. What is hard is finding the one choice that satisfies all n at the same time. Each equation is a constraint, and the solution is what survives all of them together.

Remark. For the system of §1.3 we already know the answer set has exactly one element, because $\mathcal{B}$ is a basis and coordinates are unique. That is a strong piece of information and it will not always be available. Systems in general can have one solution, no solution, or infinitely many, and deciding which is the case is a question in its own right. Here the basis hypothesis has settled it in advance, which makes this the cleanest possible first example of a linear system.

1.5. A worked case in $\mathbb{R}^2$

Take $n = 2$ and the basis

$$\mathbf{v}_1 = [2, 1], \quad \mathbf{v}_2 = [1, 3],$$

which is independent: if $\alpha_1[2, 1] + \alpha_2[1, 3] = \mathbf{0}$ then $2\alpha_1 + \alpha_2 = 0$ and $\alpha_1 + 3\alpha_2 = 0$; the first gives $\alpha_2 = -2\alpha_1$, and substituting into the second gives $\alpha_1 - 6\alpha_1 = -5\alpha_1 = 0$, so $\alpha_1 = \alpha_2 = 0$. Two independent vectors in $\mathbb{R}^2$, so a basis.

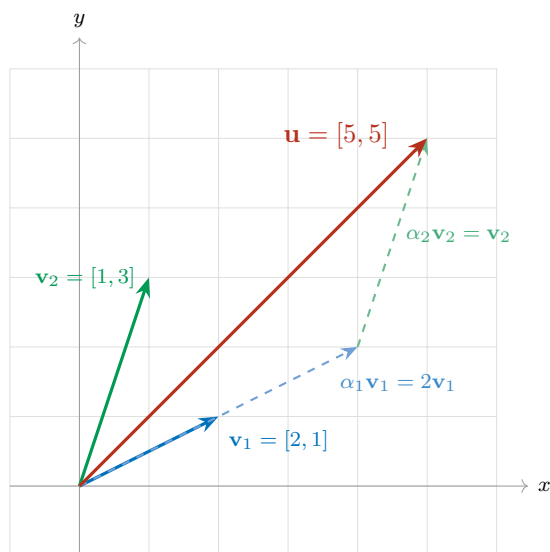
Let $\mathbf{u} = [5, 5]$ and ask for its coordinates. In the notation of §1.3, $v_{11} = 2$, $v_{21} = 1$, $v_{12} = 1$, $v_{22} = 3$, and the system is

$$2\alpha_1 + 1\alpha_2 = 5,$$

$$1\alpha_1 + 3\alpha_2 = 5.$$

Solve it by substitution: the first equation gives $\alpha_2 = 5 - 2\alpha_1$, and putting that into the second gives $\alpha_1 + 15 - 6\alpha_1 = 5$, so $-5\alpha_1 = -10$ and $\alpha_1 = 2$; then $\alpha_2 = 5 - 4 = 1$. Checking against the original vector equation,

$$2\mathbf{v}_1 + 1\mathbf{v}_2 = 2[2, 1] + [1, 3] = [4, 2] + [1, 3] = [5, 5] = \mathbf{u}. \quad \checkmark$$



The picture is the same walk as in the vector unit, §16.5 — travel α_1 steps along the first basis vector, then α_2 along the second, and arrive at the head of $\mathbf{u}$ — but the steps are no longer along the axes, and that is exactly what makes the arithmetic non-trivial.

Remark (What changed, and it is only one thing). Against the standard basis the same system would have read $1\alpha_1 + 0\alpha_2 = u_1$ and $0\alpha_1 + 1\alpha_2 = u_2$, and the zeros would have collapsed it on sight. Here there are no zeros: both unknowns appear in both equations, so neither can be found without the other, and the equations have to be handled together. The vector unit predicted this at the end of §16.5. Substitution coped with two equations; it is not a method that survives twenty.

1.6. The same thing one dimension up

One more, in $\mathbb{R}^3$, to have a case with enough room to show the shape of things. Take

$$\mathbf{v}_1 = [1, 0, 1], \quad \mathbf{v}_2 = [2, 1, 0], \quad \mathbf{v}_3 = [0, 1, 1].$$

These are independent, and checking it is itself a small system: setting $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 = \mathbf{0}$ and reading the three slots gives $\alpha_1 + 2\alpha_2 = 0$, $\alpha_2 + \alpha_3 = 0$ and $\alpha_1 + \alpha_3 = 0$. The last two give $\alpha_1 = \alpha_2 = -\alpha_3$, and substituting into the first gives $-3\alpha_3 = 0$, so all three are zero. Three independent vectors in $\mathbb{R}^3$, so a basis.

Now suppose the coordinates of some $\mathbf{u}$ turn out to be $\alpha_1 = 1, \alpha_2 = 2, \alpha_3 = 3$. Then

$$\mathbf{u} = 1[1, 0, 1] + 2[2, 1, 0] + 3[0, 1, 1] = [1, 0, 1] + [4, 2, 0] + [0, 3, 3] = [5, 5, 4].$$

Read off the components with the convention of §1.3 — v_{ki} is slot k of $\mathbf{v}_i$, so $v_{12} = 2$ is the first slot of the second vector — and the system for the unknown coordinates of $\mathbf{u} = [5, 5, 4]$ is

$$1\alpha_1 + 2\alpha_2 + 0\alpha_3 = 5,$$

$$0\alpha_1 + 1\alpha_2 + 1\alpha_3 = 5,$$

$$1\alpha_1 + 0\alpha_2 + 1\alpha_3 = 4,$$

whose solution is $\alpha_1 = 1, \alpha_2 = 2, \alpha_3 = 3$, as constructed. This is the example §2 will carry through. Notice already what §1.3 promised: the coefficients going *down* the first column are 1, 0, 1, which is $\mathbf{v}_1$; down the second, 2, 1, 0, which is $\mathbf{v}_2$; down the third, 0, 1, 1, which is $\mathbf{v}_3$.

1.7. Where else these come from

The coordinate problem is how linear systems enter *this* course, and it is a good entrance because everything about it is already understood: we know a solution exists, we know it is unique, and we know what it means. It is not the only entrance. Systems of linear equations arise whenever several linear constraints must hold at once — balancing a chemical reaction, fitting a line through data, computing currents in a circuit, allocating a fixed budget across competing demands, enforcing conservation at every node of a network.

What all of these share is not their subject matter but their *shape*: a list of unknowns, and a list of linear equations relating them. That shape is what the next section names, and naming it is what lets one method serve all of them.

2. Compressing the notation

The system of §1.3 is correct and unwieldy. Writing it out costs n lines, each with n terms, so a system in twenty unknowns fills a page before any work is done — and none of that page is doing anything a symbol could not.

Before asking *how* to solve such a system, we spend this section building notation for it. Four steps, each one small.

2.1. Step 1: every equation is a dot product

Look at a single equation of the system — row k :

$$\alpha_1 v_{k1} + \alpha_2 v_{k2} + \cdots + \alpha_n v_{kn} = u_k.$$

The left-hand side is a sum of products of paired-up numbers, which is a structure we have a name for. Set

$$\mathbf{r}_k = [v_{k1}, v_{k2}, \dots, v_{kn}], \quad \mathbf{x} = [\alpha_1, \alpha_2, \dots, \alpha_n],$$

and the left-hand side is precisely $\langle \mathbf{r}_k, \mathbf{x} \rangle$, the dot product of the vector unit, §11.3. So the system is

$$\langle \mathbf{r}_1, \mathbf{x} \rangle = u_1, \quad \langle \mathbf{r}_2, \mathbf{x} \rangle = u_2, \quad \dots, \quad \langle \mathbf{r}_n, \mathbf{x} \rangle = u_n.$$

Nothing has been computed and nothing has been assumed. We have only noticed that an operation we already have describes each line.

Example 2.1. For the $\mathbb{R}^3$ system of §1.6, the three vectors $\mathbf{r}_k$ are $[1, 2, 0]$, $[0, 1, 1]$ and $[1, 0, 1]$, and with $\mathbf{x} = [1, 2, 3]$,

$$\langle [1, 2, 0], [1, 2, 3] \rangle = 1 + 4 + 0 = 5,$$

$$\langle [0, 1, 1], [1, 2, 3] \rangle = 0 + 2 + 3 = 5,$$

$$\langle [1, 0, 1], [1, 2, 3] \rangle = 1 + 0 + 3 = 4,$$

which are the three right-hand sides 5, 5, 4. ■

Definition 2.1 (Coordinate vector). The vector $\mathbf{x} = [\alpha_1, \dots, \alpha_n]$ holding the unknown coordinates is the **coordinate vector** of $\mathbf{u}$ with respect to $\mathcal{B}$.

Remark. Two different vectors of $\mathbb{R}^n$ are now in play and it is worth keeping them apart. The vector $\mathbf{u}$ is a point of the space. The vector $\mathbf{x}$ is the recipe for building it out of $\mathcal{B}$. They are equal only when $\mathcal{B}$ is the standard basis — which is the content of the vector unit’s remark at §16.4, that the bracket notation was silently reporting coordinates in one particular basis all along.

2.2. Step 2: the matrix

The n vectors $\mathbf{r}_1, \dots, \mathbf{r}_n$ carry all the coefficients of the system, and $\mathbf{x}$ carries all the unknowns. Stack the $\mathbf{r}_k$ in order, one per line, keeping their entries aligned in columns.

Definition 2.2 (Matrix). A **matrix** is a rectangular array of numbers arranged in rows and columns. The matrix of the system of §1.3 is

$$\mathbf{V} = \begin{bmatrix} v_{11} & v_{12} & \cdots & v_{1n} \\ v_{21} & v_{22} & \cdots & v_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ v_{n1} & v_{n2} & \cdots & v_{nn} \end{bmatrix},$$

with n rows and n columns. The number v_{ki} in row k and column i is the (k, i) **entry** of $\mathbf{V}$.

This is the second object of the course, after the vector, and the convention of §1.3 is what makes it painless to write down: the two indices of v_{ki} are already the two directions of the grid, row first and column second. Nothing had to be rearranged.

Now the two readings of §1.3 become geometric facts about the array.

- **Row k of $\mathbf{V}$ is $\mathbf{r}_k$** , the coefficients of the k -th equation. Rows are equations.
- **Column i of $\mathbf{V}$ is $\mathbf{v}_i$** , the i -th basis vector, written downward. Columns are the basis.

So $\mathbf{V}$ is the basis $\mathcal{B}$ with its vectors stood up side by side:

$$\mathbf{V} = \left[\begin{array}{c|c|c|c} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{array} \right],$$

which is worth remembering as the picture of what the matrix *is*, before it is anything else.

Example 2.2. The $\mathbb{R}^3$ system of §1.6 has

$$\mathbf{V} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Its rows $[1, 2, 0]$, $[0, 1, 1]$, $[1, 0, 1]$ are the three equations. Its columns, read downward, are $[1, 0, 1]$, $[2, 1, 0]$, $[0, 1, 1]$ — exactly $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. ■

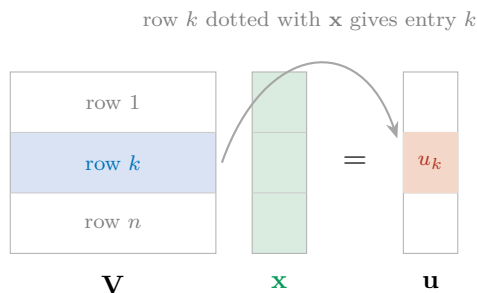
2.3. Step 3: multiplying a matrix by a vector

We have a matrix holding the coefficients and a vector holding the unknowns, and §2.1 said what to do with them: dot each row against $\mathbf{x}$. Doing that for every row produces n numbers, which is a vector. So the operation takes a matrix and a vector and returns a vector, and it deserves a name and a symbol.

Definition 2.3 (Matrix–vector product). Let $\mathbf{V}$ be an $n \times n$ matrix with rows $\mathbf{r}_1, \dots, \mathbf{r}_n$, and let $\mathbf{x} \in \mathbb{R}^n$. The **matrix–vector product** $\mathbf{V}\mathbf{x}$ is the vector of $\mathbb{R}^n$ whose k -th component is the dot product of row k with $\mathbf{x}$:

$$\mathbf{V}\mathbf{x} = \begin{bmatrix} \langle \mathbf{r}_1, \mathbf{x} \rangle \\ \langle \mathbf{r}_2, \mathbf{x} \rangle \\ \vdots \\ \langle \mathbf{r}_n, \mathbf{x} \rangle \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n v_{1i} \alpha_i \\ \sum_{i=1}^n v_{2i} \alpha_i \\ \vdots \\ \sum_{i=1}^n v_{ni} \alpha_i \end{bmatrix}.$$

In words, and this is the sentence to memorise: *to multiply a matrix by a vector, take the dot product of each row of the matrix with that vector*. The row supplies one component of the answer, and there are as many components as rows.



A common confusion (Rows, columns, and the bracket notation). The vector unit wrote vectors horizontally, $\mathbf{v} = [v_1, \dots, v_n]$, and the display above writes $\mathbf{x}$ and $\mathbf{V}\mathbf{x}$ vertically. Nothing has changed about the vectors. The bracket notation never claimed a shape — it listed n numbers in order, which is all a vector in $\mathbb{R}^n$ is — and $[1, 2, 3]$ and its column spelling denote the same element of $\mathbb{R}^3$.

The reason for switching is that $\mathbf{V}\mathbf{x}$ dots $\mathbf{x}$ against each *row*, so drawing $\mathbf{x}$ as a column puts each of its entries directly beneath the column of $\mathbf{V}$ it multiplies, and the arithmetic can be read off the page. Horizontal brackets stay in use for prose, where they take less room. When the distinction between a row and a column starts to carry mathematical weight, we will say so; here it is typography.

Example 2.3. With $\mathbf{V}$ from §2.2 and $\mathbf{x} = [1, 2, 3]$:

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1(1) + 2(2) + 0(3) \\ 0(1) + 1(2) + 1(3) \\ 1(1) + 0(2) + 1(3) \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix}.$$

Three dot products, three components. ■

Remark (A shape rule, for later). Row k has n entries and $\mathbf{x}$ has n entries, and a dot product is defined only when both vectors have the same dimension (the vector unit, §11.3). So $\mathbf{V}\mathbf{x}$ makes sense only when the number of *columns* of $\mathbf{V}$ matches the number of entries of $\mathbf{x}$. Every matrix in this part is $n \times n$ and the condition is automatic, but it is the rule that will govern all matrix multiplication later, and it comes from nowhere more exotic than the dot product needing two vectors of equal length.

2.4. Step 4: $\mathbf{V}\mathbf{x} = \mathbf{u}$

Now put the three pieces together. By §2.3, the k -th component of $\mathbf{V}\mathbf{x}$ is $\langle \mathbf{r}_k, \mathbf{x} \rangle$. By §2.1, the k -th equation of the system says that this equals u_k . Two vectors are equal exactly when all their components agree. So the entire system — all n equations, all n unknowns — is the single statement

$$\boxed{\mathbf{V}\mathbf{x} = \mathbf{u}}$$

That is the compression. The $\mathbb{R}^3$ example of §1.6, which took three lines and nine coefficients to state, is

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix},$$

and in $\mathbb{R}^{20}$ it would still be $\mathbf{V}\mathbf{x} = \mathbf{u}$.

Symbol	What it is	Known?
$\mathbf{V}$	the $n \times n$ matrix whose columns are the basis vectors	known
$\mathbf{x}$	the coordinate vector $[\alpha_1, \dots, \alpha_n]$	unknown
$\mathbf{u}$	the vector whose coordinates we want	known

Compare the shape of $\mathbf{V}\mathbf{x} = \mathbf{u}$ with the scalar equation $ax = b$, whose solution is $x = b/a$ whenever $a \neq 0$. The resemblance is not accidental and it is not yet a method: we have no division of vectors, and no meaning for $\mathbf{V}^{-1}$. But the resemblance is the reason the notation is worth having. It makes a question of n equations look like a question about one object, and questions about one object are the kind we know how to ask.

Remark (What was actually gained). Three things, and only the first is about saving ink.

Brevity. One symbol for n^2 coefficients.

Uniformity. The coordinate problem, a circuit, a least-squares fit and a network flow all reduce to $\mathbf{V}\mathbf{x} = \mathbf{u}$ with different $\mathbf{V}$. A method for the equation is a method for all of them at once.

A new object. $\mathbf{V}$ was introduced as bookkeeping, but it will not stay bookkeeping. Matrices add, multiply, and have their own arithmetic; a matrix will turn out to *act* on vectors rather than merely tabulate numbers, and $\mathbf{V}\mathbf{x}$ will be read as a function applied to $\mathbf{x}$. That reading is where most of the rest of this course lives.

2.5. The same equation, read by columns

The definition in §2.3 built $\mathbf{V}\mathbf{x}$ out of rows. There is a second way to read the same product, and it closes the loop back to §1.1.

Take the $\mathbb{R}^3$ computation again, and instead of collecting the arithmetic into three dot products, group it by which column of $\mathbf{V}$ each term came from:

$$\begin{bmatrix} 1(1) + 2(2) + 0(3) \\ 0(1) + 1(2) + 1(3) \\ 1(1) + 0(2) + 1(3) \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

The left side is grouped by row, the right side by column, and both are the same nine products added in a different order. But the right-hand side is $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3$ — a linear combination of the basis vectors, with the entries of $\mathbf{x}$ as the weights.

Theorem 2.1 (The two readings of a matrix–vector product). *Let $\mathbf{V}$ have columns $\mathbf{v}_1, \dots, \mathbf{v}_n$ and let $\mathbf{x} = [\alpha_1, \dots, \alpha_n]$. Then*

$$\mathbf{V}\mathbf{x} = \underbrace{[\langle \mathbf{r}_1, \mathbf{x} \rangle, \dots, \langle \mathbf{r}_n, \mathbf{x} \rangle]}_{\text{by rows}} = \underbrace{\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n}_{\text{by columns}}.$$

Proof. Compare slot k of each side. On the left it is $\langle \mathbf{r}_k, \mathbf{x} \rangle = \sum_{i=1}^n v_{ki} \alpha_i$, by §2.3. On the right, slot k of $\alpha_i \mathbf{v}_i$ is $\alpha_i v_{ki}$, since scaling acts componentwise, and adding the n vectors adds their k -th slots, giving $\sum_{i=1}^n \alpha_i v_{ki}$. The two sums have the same terms. $\square$

So $\mathbf{V}\mathbf{x} = \mathbf{u}$ says, in the column reading,

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n = \mathbf{u},$$

which is the boxed equation of §1.1 — the one we started from. We have gone around a full circle and arrived back where we began, in new notation.

Remark (Why the circle was worth walking). Nothing was proved on the trip; the two ends are the same statement. What changed is that the statement now has a *form*, and the form is what can be worked on. “Find weights that combine these vectors into $\mathbf{u}$ ” is a description. “Solve $\mathbf{V}\mathbf{x} = \mathbf{u}$ ” is a problem with an object in it.

Keep both readings. The row reading is how you *compute* a matrix–vector product, and it is what an algorithm implements. The column reading is how you *understand* one: $\mathbf{V}\mathbf{x}$ is a linear combination of the columns of $\mathbf{V}$. Most of the structural theorems ahead — when a system has a solution, how many, what the columns of a matrix span — are consequences of the column reading.

A common confusion ($\mathbf{V}$ is not a list of the basis vectors). Writing $\mathbf{V}$ as its columns is a picture, not an identification. A basis is a *set* — $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, unordered, with no repeats. A matrix is an *array*, in which order matters and entries may repeat. Swapping two columns of $\mathbf{V}$ gives a different matrix but the same basis, with the coordinates permuted to match. So $\mathbf{V}$ records the basis *together with* a choice of ordering for it, which the set $\mathcal{B}$ alone does not supply.

2.6. Practice

P1. Let $\mathbf{v}_1 = [3, 1]$ and $\mathbf{v}_2 = [1, 2]$, and let $\mathbf{u} = [7, 4]$. Write the system for the coordinates of $\mathbf{u}$, solve it, and check your answer in the original vector equation.

P2. Write the matrix $\mathbf{V}$ for the basis in P1, and verify that $\mathbf{V}\mathbf{x} = \mathbf{u}$ for the $\mathbf{x}$ you found.

P3. Compute $\begin{bmatrix} 2 & -1 & 0 \\ 1 & 1 & 3 \\ 0 & 4 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ twice: once by the row reading, once by the column reading. The answers must agree.

P4. Let $\mathcal{B} = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ be the standard basis of $\mathbb{R}^n$. Write down $\mathbf{V}$, and use §2.5 to say in one sentence why the system $\mathbf{V}\mathbf{x} = \mathbf{u}$ has solution $\mathbf{x} = \mathbf{u}$.

P5. Suppose $\mathbf{v}_1 = [1, 2]$ and $\mathbf{v}_2 = [2, 4]$. Show that the system for the coordinates of $\mathbf{u} = [3, 1]$ has no solution, and say which hypothesis of §1.1 fails.

Answer key

P1. The system is $3\alpha_1 + \alpha_2 = 7$ and $\alpha_1 + 2\alpha_2 = 4$. From the first, $\alpha_2 = 7 - 3\alpha_1$; substituting, $\alpha_1 + 14 - 6\alpha_1 = 4$, so $-5\alpha_1 = -10$ and $\alpha_1 = 2$, then $\alpha_2 = 1$. Check: $2[3, 1] + 1[1, 2] = [6, 2] + [1, 2] = [7, 4]$. ✓

P2. The columns of $\mathbf{V}$ are the basis vectors, so $\mathbf{V} = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ and $\mathbf{x} = [2, 1]$. Row 1: $3(2) + 1(1) = 7$. Row 2: $1(2) + 2(1) = 4$. So $\mathbf{V}\mathbf{x} = [7, 4] = \mathbf{u}$. ✓

P3. By rows: $2(1) - 1(2) + 0(-1) = 0$; $1(1) + 1(2) + 3(-1) = 0$; $0(1) + 4(2) - 2(-1) = 10$. So the product is $[0, 0, 10]$. By columns:

$$1 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix} + (-1) \begin{bmatrix} 0 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 2 \\ 8 \end{bmatrix} + \begin{bmatrix} 0 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix}. \quad \checkmark$$

P4. The columns are $\mathbf{e}_1, \dots, \mathbf{e}_n$, so $\mathbf{V}$ has 1s down the diagonal and 0s elsewhere. By the column reading, $\mathbf{V}\mathbf{x} = \alpha_1\mathbf{e}_1 + \dots + \alpha_n\mathbf{e}_n = [\alpha_1, \dots, \alpha_n] = \mathbf{x}$, so the equation reads $\mathbf{x} = \mathbf{u}$ directly. This is the vector unit's observation at §24.1 that coordinates against the standard basis are the components themselves, now visible as a property of one particular matrix.

P5. The system is $\alpha_1 + 2\alpha_2 = 3$ and $2\alpha_1 + 4\alpha_2 = 1$. The second equation is twice the left-hand side of the first, so it demands $2(3) = 1$, which is false; no pair (α_1, α_2) satisfies both. The hypothesis that fails is *independence*: $\mathbf{v}_2 = 2\mathbf{v}_1$, so $\mathcal{B}$ is linearly dependent (the vector unit, §18.4), hence not a basis, and its span is a line rather than $\mathbb{R}^2$ — a line that does not contain $[3, 1]$. With the basis hypothesis gone, so is the guarantee of §1.1 that a solution exists at all.

Looking ahead. We can now state the problem in one line but not yet solve it. Before a method for solving it is worth writing down, the object it operates on deserves a proper definition. So far $\mathbf{V}$ has been *a* matrix — the one that happened to fall out of one particular problem, with the basis vectors down its columns. The rest of this document steps back from that problem entirely and asks what a matrix is in general, what shapes they come in, and what you are allowed to do to them.

3. What a matrix is

3.1. The definition

Definition 3.1 (Matrix). Let $m, n \in \mathbb{N}$. An $m \times n$ **matrix** (read “ m by n ”) is a two-dimensional grid of numbers arranged in m rows and n columns. The numbers m and n are its **dimensions**. Each number in the grid is an **entry**, and the entry in row i and column j is written a_{ij} .

Written out, a matrix $\mathbf{A}$ is

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}.$$

Read an index in two steps: a_{ij} sits in *row* i , then *column* j . Row first, column second — always. So a_{23} is in the second row and third column, and a_{32} is in the third row and second column; these are different positions and in general different numbers. Every entry has exactly one such address, and every address names exactly one entry, which is what makes the grid an unambiguous way of storing mn numbers.

From the vector unit. This is the same convention §1.3 fixed for the basis vectors, where v_{ki} meant component k of vector i and sat in row k , column i of $\mathbf{V}$. Nothing has changed; it is now stated for a matrix in general rather than for that one matrix.

A common confusion (2×3 is not 3×2). The dimensions are an *ordered* pair and the order is rows-then-columns. A 2×3 matrix is short and wide — two rows, three columns. A 3×2 matrix is tall and narrow. Reversing them describes a different object, and in §8 it will be the difference between a product that is defined and one that is not.

3.2. Where the entries come from

The definition said “numbers” without saying which. The entries are drawn from some agreed numerical set: the real numbers $\mathbb{R}$, the integers $\mathbb{Z}$, the complex numbers $\mathbb{C}$, and others besides. A set of numbers in which you can add, subtract, multiply, and divide, and have the familiar rules of arithmetic hold, is called a **field**. Both $\mathbb{R}$ and $\mathbb{C}$ are fields. The formal definition is beyond this course; we need the word only to say where the entries live.

For the remainder of this document, every matrix entry is a real number.

Once the entry set is fixed, we can name the collection of all matrices of a given shape.

Definition 3.2 (The set $\mathbb{R}^{m \times n}$). $\mathbb{R}^{m \times n}$ denotes the set of all $m \times n$ matrices whose entries are real numbers. Writing $\mathbf{A} \in \mathbb{R}^{m \times n}$ says that $\mathbf{A}$ is an $m \times n$ matrix holding mn real scalars.

The notation is deliberately parallel to $\mathbb{R}^n$ from the vector unit’s §24.1: there, n real numbers in a list; here, $m \times n$ real numbers in a grid. In both cases the superscript records the shape.

A common confusion ($\mathbb{R}^{m \times n}$ is not a field). $\mathbb{R}$ is a field. $\mathbb{R}^{m \times n}$ is not, and the notation is not claiming it is — it is only the *name of a set*, exactly as $\mathbb{R}^n$ was. The difference is easy to state even now: in a field you can divide by any nonzero element, and there is no general way to divide by a matrix. We will see in §8 that matrices fail an even more basic test, since $\mathbf{AB}$ and $\mathbf{BA}$ need not be equal.

As §2.2 already noted, capital letters conventionally denote matrices, lower case denotes vectors and scalars. That convention is why $\mathbf{A} \in \mathbb{R}^{m \times n}$ can be read at a glance.

3.3. Examples

Example 3.1. Three matrices from three different sets.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{2 \times 2}, \quad \mathbf{B} = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 6 & 7 \end{bmatrix} \in \mathbb{R}^{3 \times 2}, \quad \mathbf{D} = \begin{bmatrix} \pi & e & \pi & e \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix} \in \mathbb{R}^{2 \times 4}.$$

Reading entries out of them:

$$a_{12} = 0, \quad b_{32} = 7, \quad b_{21} = 4, \quad d_{14} = e, \quad d_{23} = \frac{1}{4}.$$

Check b_{32} against the grid: third row, second column — the row is $[6, 7]$ and its second entry is 7. And note that $\mathbf{B}$ has no entry b_{23} : it has only two columns, so the address does not exist.

Remark. $\mathbf{A}$ above is a matrix you have already met. Its columns are $\mathbf{e}_1$ and $\mathbf{e}_2$, the standard basis of $\mathbb{R}^2$, so it is the matrix that appeared in P4 of §2.6 — the one for which $\mathbf{V}\mathbf{x} = \mathbf{x}$. It gets a name in §4.3.

3.4. A vector is a matrix

Nothing in the definition forbids $m = 1$ or $n = 1$. Those two cases should look familiar.

If $\mathbf{A} \in \mathbb{R}^{1 \times n}$ then $\mathbf{A}$ has one row and n columns:

$$\mathbf{A} = [a_{11} \ a_{12} \ a_{13} \ \cdots \ a_{1n}],$$

which is a list of n numbers — a vector. Laid out this way it is called a **row vector**. If instead $\mathbf{A} \in \mathbb{R}^{m \times 1}$, there is one column and m rows:

$$\mathbf{A} = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ \vdots \\ a_{m1} \end{bmatrix},$$

a **column vector**. Both hold the same kind of data as a vector in the sense of the vector unit; they differ only in whether that data is written across or down.

From the vector unit. §2.3 flagged this and set it aside: the bracket notation $[v_1, \dots, v_n]$ never asserted a shape, and the row and column spellings denote the same vector. Now that matrices are defined in general, the two spellings are genuinely different *matrices* — one is $1 \times n$, the other $n \times 1$ — even though they carry identical numbers. §6.3 gives the operation that converts between them.

3.5. Rows and columns

Because a single row is a row vector and a single column is a column vector, an $m \times n$ matrix admits two readings, and both are useful.

Reading	What the matrix is
By rows	A stack of m row vectors, each with n components, laid one on top of the next.
By columns	A shelf of n column vectors, each with m components, standing side by side.

For $\mathbf{B} \in \mathbb{R}^{3 \times 2}$ above, the row reading gives three vectors in $\mathbb{R}^2$ — $[2, 3]$, $[4, 5]$, $[6, 7]$ — and the column reading gives two vectors in $\mathbb{R}^3$ — $[2, 4, 6]$ and $[3, 5, 7]$. Same six numbers, grouped two different ways.

From the vector unit. Both readings have already done work. §2.4 computed $\mathbf{V}\mathbf{x}$ by rows, as n dot products; §2.5 computed the same product by columns, as one linear combination of the basis vectors. That the two agree was the point of §2.5, and it is a fact about every matrix, not just $\mathbf{V}$.

4. Shapes and patterns

Matrices are classified first by shape — the relationship between m and n — and then by where their nonzero entries sit. The vocabulary below is used constantly, because knowing that a matrix has a particular pattern usually means there is a faster way to work with it.

4.1. Square and rectangular

Definition 4.1. A matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, with as many rows as columns, is **square**. A matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ with $m \neq n$ is **rectangular**.

The distinction matters more than it looks. A square matrix maps $\mathbb{R}^n$ to itself and can be multiplied by itself; a rectangular one cannot. The coefficient matrix $\mathbf{V}$ of §2.2 was square precisely because a basis of $\mathbb{R}^n$ has exactly n vectors, each with n components — and that squareness is what made “ n equations in n unknowns” possible.

$$\underbrace{\begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}}_{2 \times 2, \text{ square}} \quad \underbrace{\begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 6 & 7 \end{bmatrix}}_{3 \times 2, \text{ rectangular}} \quad \underbrace{\begin{bmatrix} 1 & 0 & 4 & 2 \\ 7 & 1 & 0 & 3 \end{bmatrix}}_{2 \times 4, \text{ rectangular}}$$

Remark (The 1×1 case). A 1×1 matrix $[a_{11}]$ is technically square, but it holds a single number and behaves like one. It is normally read as the scalar a_{11} rather than treated as a matrix at all. This is not a quibble — §7.1 depends on it.

4.2. The diagonal

The entries $a_{11}, a_{22}, a_{33}, \dots$ — those with equal row and column index — form the **main diagonal**, running from the top-left corner downward. Everything else is *off-diagonal*.

Definition 4.2. A square matrix is **diagonal** if every off-diagonal entry is zero: $a_{ij} = 0$ whenever $i \neq j$.

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix} \quad \text{is diagonal, and is written } \text{diag}(3, -1, 5).$$

A diagonal matrix is described completely by n numbers rather than n^2 , and almost everything one does with matrices becomes easy when they are diagonal — a recurring theme later in the course.

4.3. The identity

Definition 4.3. The **identity matrix** $\mathbf{I}_n \in \mathbb{R}^{n \times n}$ is the diagonal matrix whose diagonal entries are all 1:

$$(\mathbf{I}_n)_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad \mathbf{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The subscript is dropped when the size is clear.

Its columns are exactly $\mathbf{e}_1, \dots, \mathbf{e}_n$, so $\mathbf{I}$ is the matrix of the standard basis — which is why P4 of §2.6 found $\mathbf{I}\mathbf{x} = \mathbf{x}$. §8.4 states the corresponding fact for matrices.

4.4. Triangular

Definition 4.4. A square matrix is **upper triangular** if every entry below the main diagonal is zero ($a_{ij} = 0$ for $i > j$), and **lower triangular** if every entry above it is zero ($a_{ij} = 0$ for $i < j$).

$$\underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}}_{\text{upper triangular}} \qquad \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}}_{\text{lower triangular}}$$

A matrix that is both upper and lower triangular has zeros above *and* below the diagonal — that is, it is diagonal.

Looking ahead. Triangular matrices are not a curiosity. A triangular system of equations can be solved by reading off one unknown at a time and substituting backwards, with no simultaneous juggling at all. Turning a general system into a triangular one is exactly what the solution method announced at the end of this document does.

4.5. Banded

Definition 4.5. A matrix is **banded** if its nonzero entries are confined to a band of diagonals around the main diagonal — that is, $a_{ij} = 0$ whenever $|i - j|$ exceeds some small fixed width.

The narrowest interesting case is **tridiagonal**: nonzeros only on the main diagonal and the two diagonals adjacent to it, so $a_{ij} = 0$ when $|i - j| > 1$.

$$\begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

Diagonal matrices are the case $|i - j| > 0$, so “diagonal” is the narrowest band of all. Bands of width 1 and 2 turn up constantly in problems where each quantity interacts only with its immediate neighbours.

4.6. Sparse and dense

Definition 4.6. A matrix is **sparse** if most of its entries are zero, and **dense** if most are nonzero.

Unlike the previous definitions this one is deliberately informal — there is no fixed fraction at which a matrix becomes sparse. The working test is practical: a matrix is sparse if it pays to store and process only the nonzero entries and ignore the rest.

$$\underbrace{\begin{bmatrix} 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 7 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \end{bmatrix}}_{\text{sparse: 5 nonzeros out of 25}} \qquad \underbrace{\begin{bmatrix} 3 & 1 & 4 \\ 1 & 5 & 9 \\ 2 & 6 & 5 \end{bmatrix}}_{\text{dense: no zeros at all}}$$

Diagonal, triangular and banded matrices are all sparse, and their sparsity has a *known shape* — you can say in advance which entries are zero. General sparse matrices need not have any pattern; they

are simply mostly empty. Real problems produce enormous sparse matrices routinely: a matrix with a million rows and only a handful of nonzeros per row is entirely ordinary, and storing it as a full grid would be hopeless.

4.7. Summary

Name	Condition	In words
Square	$m = n$	as many rows as columns
Rectangular	$m \neq n$	more of one than the other
Diagonal	$a_{ij} = 0$ for $i \neq j$	only the main diagonal survives
Identity	diagonal, all $a_{ii} = 1$	the diagonal is all ones
Upper triangular	$a_{ij} = 0$ for $i > j$	nothing below the diagonal
Lower triangular	$a_{ij} = 0$ for $i < j$	nothing above the diagonal
Banded	$a_{ij} = 0$ for $ i - j > w$	confined near the diagonal
Sparse	mostly zeros	few enough nonzeros to exploit
Dense	mostly nonzeros	no structure to exploit

5. Adding matrices

5.1. The definition

Definition 5.1 (Matrix addition). Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$ — *the same shape*. Their **sum** $\mathbf{A} + \mathbf{B} \in \mathbb{R}^{m \times n}$ is formed entrywise:

$$(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij} \quad \text{for every } i, j.$$

Subtraction is defined the same way, $(\mathbf{A} - \mathbf{B})_{ij} = a_{ij} - b_{ij}$.

Each entry minds its own business: the entry in row i , column j of the answer depends only on the entries in that same position of the two inputs. Nothing mixes across positions.

A common confusion (The shapes must match). $\mathbf{A} + \mathbf{B}$ is *undefined* unless $\mathbf{A}$ and $\mathbf{B}$ have the same number of rows *and* the same number of columns. There is no way to add a 3×2 to a 2×2 : some entries would have no partner. This is the same restriction the vector unit placed on adding vectors (§7.1) — you cannot add a vector in $\mathbb{R}^2$ to one in $\mathbb{R}^3$ — and for the same reason.

5.2. Examples

Example 5.1. Three sums and one non-sum.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}, \quad \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 6 & 7 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ -6 & -7 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 5 & 5 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 6 & 7 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{is undefined: } 3 \times 2 \text{ against } 2 \times 2.$$

5.3. Scaling a matrix

One more operation is needed before the laws can be stated. For a scalar $c \in \mathbb{R}$ and $\mathbf{A} \in \mathbb{R}^{m \times n}$, the **scalar multiple** $c\mathbf{A}$ multiplies every entry by c :

$$(c\mathbf{A})_{ij} = c a_{ij}, \quad \text{for example} \quad 3 \begin{bmatrix} 1 & -2 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 3 & -6 \\ 0 & 12 \end{bmatrix}.$$

In particular $-\mathbf{A} = (-1)\mathbf{A}$ negates every entry, and $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$ — subtraction is addition of the negative, exactly as it was for vectors (§7.3 of the vector unit).

The **zero matrix** $\mathbf{0} \in \mathbb{R}^{m \times n}$ has every entry equal to 0.

5.4. The algebraic laws

Because addition is entrywise, every law is inherited from the arithmetic of $\mathbb{R}$ one entry at a time. Each proof is the same sentence: *compare the (i, j) entries of both sides and apply the corresponding law for real numbers.*

	Law	Name
A1	$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$	commutativity
A2	$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$	associativity
A3	$\mathbf{A} + \mathbf{0} = \mathbf{A}$	additive identity
A4	$\mathbf{A} + (-\mathbf{A}) = \mathbf{0}$	additive inverse
S1	$c(\mathbf{A} + \mathbf{B}) = c\mathbf{A} + c\mathbf{B}$	distributivity over matrix sums
S2	$(c + d)\mathbf{A} = c\mathbf{A} + d\mathbf{A}$	distributivity over scalar sums
S3	$c(d\mathbf{A}) = (cd)\mathbf{A}$	compatibility of scaling
S4	$1\mathbf{A} = \mathbf{A}$	scaling identity

To see A1 in full: the (i, j) entry of $\mathbf{A} + \mathbf{B}$ is $a_{ij} + b_{ij}$, the (i, j) entry of $\mathbf{B} + \mathbf{A}$ is $b_{ij} + a_{ij}$, and these are equal because addition of *real numbers* commutes. Since the two matrices agree in every position, they are the same matrix. The rest go identically.

Remark. This is the identical list the vector unit proved in §8.4 and §11.8, with matrices in place of vectors. That is not a coincidence, and it is worth noticing: a set whose elements can be added and scaled subject to exactly these eight laws is what the course will eventually call a *vector space*. By that standard $\mathbb{R}^{m \times n}$ qualifies — matrices are, for the purposes of addition and scaling, vectors with a rectangular filing system. What separates them arrives in §8.

6. The transpose

6.1. The definition

Definition 6.1 (Transpose). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Its **transpose** $\mathbf{A}^T \in \mathbb{R}^{n \times m}$ is the matrix whose (i, j) entry is the (j, i) entry of $\mathbf{A}$:

$$(\mathbf{A}^T)_{ij} = a_{ji}.$$

Two things happen at once. The indices swap, so the entry at row i , column j comes from row j , column i ; and consequently the *shape* swaps, an $m \times n$ matrix becoming $n \times m$.

Geometrically it is a reflection across the main diagonal. The diagonal entries a_{ii} stay where they are — swapping i with i changes nothing — and everything else is mirrored through that line. Operationally, the cleanest way to compute it: **the rows of $\mathbf{A}$ become the columns of $\mathbf{A}^\top$** , in order.

6.2. Examples

Example 6.1. Taking $\mathbf{B}$ from §3.3:

$$\mathbf{B} = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 6 & 7 \end{bmatrix} \in \mathbb{R}^{3 \times 2}, \quad \mathbf{B}^\top = \begin{bmatrix} 2 & 4 & 6 \\ 3 & 5 & 7 \end{bmatrix} \in \mathbb{R}^{2 \times 3}.$$

The rows of $\mathbf{B}$ — $[2, 3]$, $[4, 5]$, $[6, 7]$ — are the columns of $\mathbf{B}^\top$. Checking one entry: $(\mathbf{B}^\top)_{12} = b_{21} = 4$. ✓

A square example, where the shape does not change but the entries still move:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}^\top = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}.$$

The diagonal 1, 5, 9 is fixed; every other entry has traded places with its mirror image.

6.3. Row and column vectors

The transpose is precisely the operation §3.4 promised. A vector $\mathbf{v} \in \mathbb{R}^n$ written as a column is an $n \times 1$ matrix; transposing it gives a $1 \times n$ matrix, which is the same vector written as a row:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^{n \times 1}, \quad \mathbf{v}^\top = [v_1 \quad v_2 \quad \cdots \quad v_n] \in \mathbb{R}^{1 \times n}.$$

Convention. From here on, an unadorned vector $\mathbf{v}$ means a *column* vector, and $\mathbf{v}^\top$ is its row form. This settles the ambiguity §2.3 set aside: the bracket notation $[v_1, \dots, v_n]$ remains a convenient way to write a vector inline, but whenever shape matters, $\mathbf{v}$ is $n \times 1$.

6.4. Properties

Law	In words
T1 $(\mathbf{A}^\top)^\top = \mathbf{A}$	transposing twice returns the original
T2 $(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$	transpose distributes over addition
T3 $(c\mathbf{A})^\top = c\mathbf{A}^\top$	scalars pass through

Each is an entry comparison. For T1, the (i, j) entry of $(\mathbf{A}^\top)^\top$ is the (j, i) entry of $\mathbf{A}^\top$, which is the (i, j) entry of $\mathbf{A}$; the indices swap twice and land back where they started.

Looking ahead. There is a fourth law, for the transpose of a *product*, and it contains a surprise. It has to wait for §8.4, since products are not defined yet.

6.5. Symmetric matrices

Definition 6.2. A matrix $\mathbf{A}$ is **symmetric** if $\mathbf{A}^\top = \mathbf{A}$, that is, if $a_{ij} = a_{ji}$ for every i and j .

A symmetric matrix must be square: $\mathbf{A}^\top$ has shape $n \times m$ and $\mathbf{A}$ has shape $m \times n$, so they cannot be equal unless $m = n$. Beyond that, symmetry says the matrix is unchanged by reflection across its main diagonal — whatever sits above the diagonal is mirrored below it.

$$\mathbf{S} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 6 \end{bmatrix} \quad \text{is symmetric:} \quad s_{12} = s_{21} = 2, \quad s_{13} = s_{31} = 3, \quad s_{23} = s_{32} = 4.$$

The diagonal entries are unconstrained; only the off-diagonal pairs must agree. A symmetric $n \times n$ matrix is therefore determined by $n(n+1)/2$ numbers rather than n^2 — the diagonal plus one triangle.

Remark. Diagonal matrices are symmetric, since all their off-diagonal entries are 0 and $0 = 0$. A triangular matrix that is symmetric must be diagonal.

7. Inner and outer products

The transpose lets us write something we already know in a new way — and the new way turns out to generalise.

7.1. The dot product as $\mathbf{v}^\top \mathbf{u}$

Take $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, both columns, so both $n \times 1$. Then $\mathbf{v}^\top$ is $1 \times n$. Place them side by side and multiply row-by-column, the way §2.3 multiplied a row of $\mathbf{V}$ into $\mathbf{x}$:

$$\mathbf{v}^\top \mathbf{u} = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} = v_1 u_1 + v_2 u_2 + \cdots + v_n u_n = \sum_{k=1}^n v_k u_k.$$

That sum is the dot product $\langle \mathbf{v}, \mathbf{u} \rangle$ of the vector unit's §11.3, unchanged. What is new is the packaging.

Why the result is a scalar. Track the shapes. A $1 \times n$ against an $n \times 1$ produces a matrix with 1 row and 1 column — a 1×1 matrix. And by the remark in §4.1, a 1×1 matrix holds a single number and is read as that number. So

$$\mathbf{v}^\top \mathbf{u} \in \mathbb{R}^{1 \times 1} = \text{a scalar},$$

which is exactly what the dot product is required to be. The notation and the definition agree. Written this way the dot product is called the **inner product**, and $\mathbf{v}^\top \mathbf{u}$ is the standard way to write it from here on.

Example 7.1. For $\mathbf{v} = [1, 2, 3]$ and $\mathbf{u} = [4, 5, 6]$,

$$\mathbf{v}^\top \mathbf{u} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = 4 + 10 + 18 = 32.$$

One number, as promised.

7.2. The outer product

Now reverse the order. If $\mathbf{v}^\top \mathbf{u}$ is a $1 \times n$ times an $n \times 1$, then $\mathbf{u}\mathbf{v}^\top$ is an $n \times 1$ times a $1 \times n$ — and the shapes say the result is $n \times n$. This is a different object entirely.

Definition 7.1 (Outer product). For $\mathbf{u} \in \mathbb{R}^m$ and $\mathbf{v} \in \mathbb{R}^n$, the **outer product** is the $m \times n$ matrix $\mathbf{u}\mathbf{v}^\top$ whose (i, j) entry is $u_i v_j$:

$$\mathbf{u}\mathbf{v}^\top = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix} \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} = \begin{bmatrix} u_1 v_1 & u_1 v_2 & \cdots & u_1 v_n \\ u_2 v_1 & u_2 v_2 & \cdots & u_2 v_n \\ \vdots & \vdots & \ddots & \vdots \\ u_m v_1 & u_m v_2 & \cdots & u_m v_n \end{bmatrix}.$$

Note what the definition does *not* require: $\mathbf{u}$ and $\mathbf{v}$ need not have the same number of components. The inner product demands it — the sum $\sum_k v_k u_k$ is meaningless otherwise — but the outer product pairs every component of $\mathbf{u}$ with every component of $\mathbf{v}$, and that makes sense for any two lengths. It simply produces a rectangular matrix.

Example 7.2. With $\mathbf{u} = [1, 2, 3] \in \mathbb{R}^3$ and $\mathbf{v} = [4, 5] \in \mathbb{R}^2$:

$$\mathbf{u}\mathbf{v}^\top = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 4 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 8 & 10 \\ 12 & 15 \end{bmatrix} \in \mathbb{R}^{3 \times 2}.$$

Here $\mathbf{v}^\top \mathbf{u}$ would be undefined — a 1×2 against a 3×1 has nothing to pair up.

A common confusion (Order changes everything). $\mathbf{v}^\top \mathbf{u}$ and $\mathbf{u}\mathbf{v}^\top$ are built from the same two vectors and are not remotely the same thing. The first is a single number; the second is a whole matrix. The transpose's position is what distinguishes them, so it is not decoration — read it carefully.

7.3. The norm, rewritten

Set $\mathbf{u} = \mathbf{v}$ in the inner product:

$$\mathbf{v}^\top \mathbf{v} = \sum_{k=1}^n v_k^2 = v_1^2 + v_2^2 + \cdots + v_n^2.$$

That sum is the square of the Euclidean norm from the vector unit's §11.9. So

$$\mathbf{v}^\top \mathbf{v} = \|\mathbf{v}\|^2, \quad \text{equivalently} \quad \|\mathbf{v}\| = \sqrt{\mathbf{v}^\top \mathbf{v}}. \quad (1)$$

The vector unit wrote this as $\langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{v}\|^2$; (1) is the same identity in transpose notation.

Example 7.3. For $\mathbf{v} = [3, 4]$: $\mathbf{v}^\top \mathbf{v} = 9 + 16 = 25$, so $\|\mathbf{v}\| = \sqrt{25} = 5$. ✓

8. Matrix multiplication

This section defines the operation that makes matrices more than rectangular storage. We give it here as a *procedure* — a rule for producing one matrix from two others. No attempt is made yet to justify why the rule is this one; that motivation comes later in the course. What can be said now is that it is among the most consequential operations in all of scientific computing, and that everything built on matrices from here forward runs through it.

8.1. The rule

You have already computed the basic step twice. §2.3 multiplied a row of $\mathbf{V}$ into a column $\mathbf{x}$; §7.1 multiplied a $1 \times n$ row into an $n \times 1$ column. Both took a row and a column of matching length and returned one number — their dot product.

Matrix multiplication is that step, performed for every row of the first matrix against every column of the second.

Definition 8.1 (Matrix multiplication). Let $\mathbf{A} \in \mathbb{R}^{m \times p}$ and $\mathbf{B} \in \mathbb{R}^{p \times n}$. Their **product** $\mathbf{C} = \mathbf{AB}$ is the $m \times n$ matrix whose (i, j) entry is the dot product of row i of $\mathbf{A}$ with column j of $\mathbf{B}$:

$$c_{ij} = \sum_{k=1}^p a_{ik} b_{kj}.$$

Read the formula as a walk: k runs along row i of $\mathbf{A}$ left-to-right while simultaneously running down column j of $\mathbf{B}$ top-to-bottom, multiplying the pair it lands on and accumulating the total. The row index i and column index j are held fixed throughout; they select *which* entry of the answer is being computed.

8.2. The dimensions must line up

The sum $\sum_{k=1}^p a_{ik} b_{kj}$ only makes sense if row i of $\mathbf{A}$ and column j of $\mathbf{B}$ have the *same length*. Row i of $\mathbf{A}$ has as many entries as $\mathbf{A}$ has columns; column j of $\mathbf{B}$ has as many entries as $\mathbf{B}$ has rows. So:

$\mathbf{AB}$ is defined exactly when the number of **columns of A** equals the number of **rows of B**.

$$(m \times \underbrace{p}_{\text{must match}}) (\underbrace{p}_{\text{must match}} \times n) \rightarrow m \times n$$

The matching pair — the two p 's — are the **inner** dimensions, and they vanish from the answer. The **outer** dimensions m and n survive and give the product its shape. Writing the shapes side by side and checking that the inner two agree is the first thing to do before any matrix multiplication, and it catches most errors before any arithmetic happens.

Example 8.1 (Dimension bookkeeping).

$$\begin{array}{llll} (2 \times 3)(3 \times 4) & \rightarrow 2 \times 4 & \text{defined} & (3 = 3) \\ (3 \times 2)(2 \times 2) & \rightarrow 3 \times 2 & \text{defined} & (2 = 2) \\ (2 \times 3)(2 \times 3) & \rightarrow \text{---} & \text{undefined} & (3 \neq 2) \\ (1 \times n)(n \times 1) & \rightarrow 1 \times 1 & \text{defined} & \text{the inner product, §7.1} \\ (m \times 1)(1 \times n) & \rightarrow m \times n & \text{defined} & \text{the outer product, §7.2} \end{array}$$

The last two lines are worth dwelling on: the inner and outer products are not special constructions at all. They are ordinary matrix multiplication applied to the two thinnest possible shapes.

8.3. Examples

Example 8.2 (Two square matrices). Let $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$, both 2×2 , so the product is 2×2 . Each entry is one dot product:

$$\mathbf{AB} = \begin{bmatrix} 1(5) + 2(7) & 1(6) + 2(8) \\ 3(5) + 4(7) & 3(6) + 4(8) \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

The entry 19 came from row 1 of $\mathbf{A}$ against column 1 of $\mathbf{B}$; the entry 43 from row 2 against column 1. Trace one of them by hand before reading on.

Example 8.3 (Rectangular). Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \in \mathbb{R}^{2 \times 3}$ and $\mathbf{B} = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} \in \mathbb{R}^{3 \times 2}$. The inner dimensions are both 3, so the product is defined and is 2×2 :

$$\mathbf{AB} = \begin{bmatrix} 1(7) + 2(9) + 3(11) & 1(8) + 2(10) + 3(12) \\ 4(7) + 5(9) + 6(11) & 4(8) + 5(10) + 6(12) \end{bmatrix} = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix}.$$

Note that $\mathbf{BA}$ is *also* defined here — $(3 \times 2)(2 \times 3)$ — but produces a 3×3 matrix. The two products are not merely different numbers; they are different sizes.

Example 8.4 (Multiplying by the identity).

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1(a) + 0(c) & 1(b) + 0(d) \\ 0(a) + 1(c) & 0(b) + 1(d) \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

The identity leaves its partner untouched, which is where it gets its name.

8.4. The algebraic laws

Law	Name
M1 $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$	associativity
M2 $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$	left distributivity
M3 $(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$	right distributivity
M4 $c(\mathbf{AB}) = (c\mathbf{A})\mathbf{B} = \mathbf{A}(c\mathbf{B})$	scalars move freely
M5 $\mathbf{IA} = \mathbf{AI} = \mathbf{A}$	multiplicative identity
M6 $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$	transpose of a product

Each assumes the shapes are compatible; M2 and M3 need two separate statements precisely *because* the sides cannot be interchanged.

M6 reverses the order. This is the surprise promised in §6.4. The shapes alone force it: if $\mathbf{A}$ is $m \times p$ and $\mathbf{B}$ is $p \times n$, then $(\mathbf{AB})^\top$ is $n \times m$. The only way to build an $n \times m$ matrix from $\mathbf{A}^\top$ ($p \times m$) and $\mathbf{B}^\top$ ($n \times p$) is $\mathbf{B}^\top \mathbf{A}^\top$ — writing $\mathbf{A}^\top \mathbf{B}^\top$ would put m against n and usually not even be defined.

A *common confusion* (Multiplication does not commute). $\mathbf{AB}$ and $\mathbf{BA}$ are in general *different*. There is no law M0 saying otherwise, and this is the single most important way matrix arithmetic departs from arithmetic with numbers. It can fail three ways, in increasing order of severity:

- $\mathbf{BA}$ may be *undefined* while $\mathbf{AB}$ is fine — try $(2 \times 3)(3 \times 4)$.
- Both may be defined but have *different shapes* — the $(2 \times 3)(3 \times 2)$ example above gives 2×2 one way and 3×3 the other.
- Both may be defined and the same shape and still differ. With $\mathbf{A}$ and $\mathbf{B}$ from the first example,

$$\mathbf{AB} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}, \quad \mathbf{BA} = \begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix}.$$

So the order in which matrices are written is part of the statement, never a stylistic choice. This is also the promised reason $\mathbb{R}^{m \times n}$ is not a field.

8.5. Matrix–vector products, revisited

A vector is a matrix (§3.4), so the matrix–vector product of §2.3 needs no separate definition — it is matrix multiplication with $n = 1$. For $\mathbf{A} \in \mathbb{R}^{m \times p}$ and $\mathbf{x} \in \mathbb{R}^p$ written as a $p \times 1$ column,

$$(m \times p)(p \times 1) \longrightarrow m \times 1,$$

so $\mathbf{Ax}$ is a column vector with m components, and its i -th component is

$$(\mathbf{Ax})_i = \sum_{k=1}^p a_{ik}x_k,$$

the dot product of row i of $\mathbf{A}$ with $\mathbf{x}$. That is precisely the definition given in §2.3, now arriving as a special case rather than as its own rule.

Example 8.5. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1(5) + 2(6) \\ 3(5) + 4(6) \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}$, a 2×1 column.

And $\mathbf{Vx} = \mathbf{u}$, the equation this document was built to write down, is now an equation between two $n \times 1$ matrices.

8.6. Practice: shapes first

For each, state whether the product is defined and give its dimensions. Compute only where indicated.

P1. $\mathbf{A} \in \mathbb{R}^{4 \times 3}$, $\mathbf{B} \in \mathbb{R}^{3 \times 5}$: is $\mathbf{AB}$ defined? Is $\mathbf{BA}$?

P2. $\mathbf{u} \in \mathbb{R}^5$, $\mathbf{v} \in \mathbb{R}^5$, both columns. Give the shapes of $\mathbf{u}^T \mathbf{v}$ and $\mathbf{uv}^T$.

P3. Compute $\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 5 & 2 \end{bmatrix}$ and $\begin{bmatrix} 1 & 4 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$.

P4. $\mathbf{A} \in \mathbb{R}^{3 \times 2}$. What are the shapes of $\mathbf{A}^T \mathbf{A}$ and $\mathbf{AA}^T$? Are they the same matrix?

P5. Compute $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$.

Answers

P1. $\mathbf{AB}$: inner dimensions 3 and 3 agree, so it is defined and is 4×5 . **$\mathbf{BA}$:** the shapes are $(3 \times 5)(4 \times 3)$, and $5 \neq 4$, so it is undefined.

P2. $\mathbf{u}^T \mathbf{v}$ is $(1 \times 5)(5 \times 1) = 1 \times 1$, a scalar — the inner product. **$\mathbf{uv}^T$** is $(5 \times 1)(1 \times 5) = 5 \times 5$, a matrix. Same two vectors, wildly different objects.

P3.

$$\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} 2+5 & 8+2 \\ 0+15 & 0+6 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 15 & 6 \end{bmatrix}, \quad \begin{bmatrix} 1 & 4 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 2+0 & 1+12 \\ 10+0 & 5+6 \end{bmatrix} = \begin{bmatrix} 2 & 13 \\ 10 & 11 \end{bmatrix}.$$

Both defined, both 2×2 , and not equal — another instance of the warning in §8.4.

P4. $\mathbf{A}^T$ is 2×3 . So **$\mathbf{A}^T \mathbf{A}$** is $(2 \times 3)(3 \times 2) = 2 \times 2$, and **$\mathbf{AA}^T$** is $(3 \times 2)(2 \times 3) = 3 \times 3$. Different shapes, so certainly not the same matrix. Both are defined for *any* $\mathbf{A}$, and by M6 both are symmetric: $(\mathbf{A}^T \mathbf{A})^T = \mathbf{A}^T (\mathbf{A}^T)^T = \mathbf{A}^T \mathbf{A}$.

P5. $(2 \times 3)(3 \times 1) = 2 \times 1$: $\begin{bmatrix} 1(2) + 2(1) + 0(4) \\ 0(2) + 1(1) + 3(4) \end{bmatrix} = \begin{bmatrix} 4 \\ 13 \end{bmatrix}$.

9. The trace

One more quantity, extracted from a matrix by ignoring almost all of it.

9.1. The definition

Definition 9.1 (Trace). Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be square. Its **trace** is the sum of its diagonal entries:

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \cdots + a_{nn}.$$

The trace is a *scalar* — one number summarising an $n \times n$ grid. It is defined only for square matrices, since a rectangular matrix has no full diagonal to sum. Every off-diagonal entry is discarded.

Example 9.1.

$$\text{tr} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = 1 + 4 = 5, \quad \text{tr} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 6 \end{bmatrix} = 1 + 5 + 6 = 12, \quad \text{tr}(\mathbf{I}_n) = \underbrace{1 + 1 + \cdots + 1}_n = n.$$

The middle matrix is the symmetric $\mathbf{S}$ of §6.5; only its diagonal mattered. The last is worth remembering: the trace of the $n \times n$ identity is n , not 1.

9.2. Properties

Law	Name
R1 $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$	additivity
R2 $\text{tr}(c\mathbf{A}) = c \text{tr}(\mathbf{A})$	scalars pass through
R3 $\text{tr}(\mathbf{A}^T) = \text{tr}(\mathbf{A})$	transpose leaves it alone
R4 $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$	the order does not matter

R1 and R2 are immediate: addition and scaling are entrywise, so they act on the diagonal entries the same way they act on all the others. R3 holds because transposing fixes the diagonal — $(\mathbf{A}^\top)_{ii} = a_{ii}$, so the sum is unchanged.

R4 is the interesting one.

9.3. Why $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$

Take $\mathbf{A} \in \mathbb{R}^{m \times p}$ and $\mathbf{B} \in \mathbb{R}^{p \times m}$, so that both products exist. Write each trace out with the definition of the product from §8.1:

$$\text{tr}(\mathbf{AB}) = \sum_{i=1}^m (\mathbf{AB})_{ii} = \sum_{i=1}^m \sum_{k=1}^p a_{ik} b_{ki}, \quad \text{tr}(\mathbf{BA}) = \sum_{k=1}^p (\mathbf{BA})_{kk} = \sum_{k=1}^p \sum_{i=1}^m b_{ki} a_{ik}.$$

These are the same double sum. Every product $a_{ik} b_{ki}$ appears exactly once on each side; the two expressions differ only in which index is summed first, and a finite sum does not care about that. Hence the two traces are equal.

Remark. Notice how strong this is. §8.4 warned that $\mathbf{AB}$ and $\mathbf{BA}$ are in general different matrices — and here they need not even be the *same size*: $\mathbf{AB}$ is $m \times m$ while $\mathbf{BA}$ is $p \times p$. Two different matrices, of two different shapes, nevertheless have identical traces.

Example 9.2. Take $\mathbf{A}$ and $\mathbf{B}$ from Example 8.3, where $\mathbf{AB} = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix}$ was computed. Its trace is $58 + 154 = 212$. Now $\mathbf{BA}$ is 3×3 , and R4 says we need only its diagonal:

$$(\mathbf{BA})_{11} = 7(1) + 8(4) = 39, \quad (\mathbf{BA})_{22} = 9(2) + 10(5) = 68, \quad (\mathbf{BA})_{33} = 11(3) + 12(6) = 105,$$

and $39 + 68 + 105 = 212$. ✓ A 2×2 matrix and a 3×3 matrix agreeing on a single number.

Remark. Setting $\mathbf{B} = \mathbf{A}^\top$ gives a useful special case. The (i, i) entry of $\mathbf{A}^\top \mathbf{A}$ is the dot product of column i of $\mathbf{A}$ with itself, so

$$\text{tr}(\mathbf{A}^\top \mathbf{A}) = \sum_i \sum_j a_{ij}^2,$$

the sum of the squares of *every* entry of $\mathbf{A}$. This is the matrix counterpart of $\mathbf{v}^\top \mathbf{v} = \|\mathbf{v}\|^2$ from §7.3, and it is one sensible way to measure the size of a matrix.

Looking ahead. Two threads are now open. First, we can state the coordinate problem as $\mathbf{V}\mathbf{x} = \mathbf{u}$ but still cannot solve it, and substitution — fine for the 2×2 case of §1.5 — is not a method that scales. What is needed is a procedure that works on $\mathbf{V}$ directly, applies to any size, and terminates with the answer or with a definite report that there is none. That procedure is *Gaussian elimination*, and it is next. With it come the questions P5 of §2.6 already raised: when does a system have a solution, when is it unique, and what does the matrix look like in each case — and §4.4 has already hinted that the answer involves making the matrix triangular.

Second, matrix multiplication was given here as a rule to follow, with its cost left uncounted. Taken as an algorithm it has a definite operation count, and that count is examined in the computational unit.